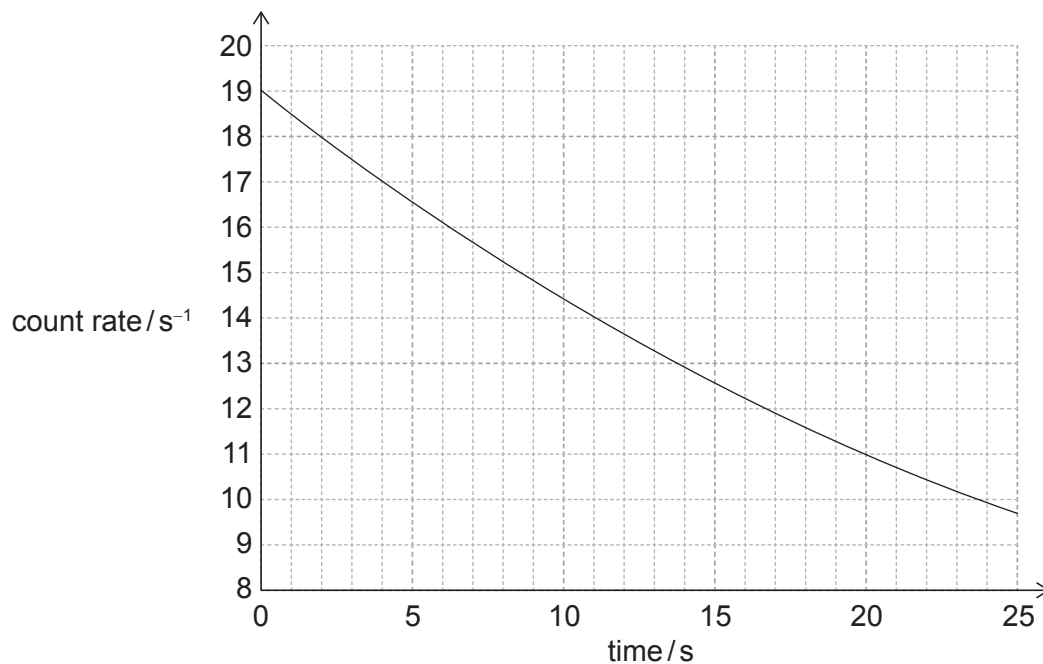


E.3 / PAPER 1 / QUESTION 25

4 November 2021 / TZ0 / HL Paper 1 / original Q20 / 8821-6501

25. A detector measures the count rate from a sample of a radioactive nuclide. The graph shows the variation with time of the count rate.

The nuclide has a half-life of 20 s. The average background count rate is constant.



What is the average background count rate?

- A. 1 s^{-1}
- B. 2 s^{-1}
- C. 3 s^{-1}
- D. 4 s^{-1}